# Notes of Database 2

*A series of not yet structured notes on the course of Database 2 as taught in Politecnico di Milano by Sara Comai during the academic year 2018/2019.*

**Notes of Database 2**
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# Concurrency Control

## Class Hierarchy

$$2PL \implies CSR \implies VSR$$

## VSR - View Serializability

***Preliminary Definitions***

- <u>*Reads-From*</u>
  $r_i(x)$ ***reads-from*** $w_j(x)$ in a schedule $S$ when $w_j(x)$ precedes $r_i(x)$ and there is no $w_k(x)$ in $S$ between $r_i(x)$ and $w_j(x)$
- <u>*Final Write*</u>
  $w_i(x)$ in a schedule $S$ is a ***final write*** if it is the last write on $x$ the occurs in $S$

***View Equivalence***
Two schedules are ***view-equivalent*** $(S_i \approx_{\mathbf{v}} S_j)$ if: they have the same

- *operations*
- *reads-from* relations
- *final writes*

***View Serializable***
A schedule is ***view-serializable*** if it is view-equivalent to a serial schedule of the same transactions

The class of View-Serializable schedules is named ***VSR***.

***Blind Writes***

a write $w_i(X)$ is said to be ***blind*** if it is not the last action of resource $X$ and the following action on $X$ is a write $w_j(X)$.

***VSR Property*** each schedule $S \in VSR$ and $S \notin CSR$ has, in its conflict graph, cycles of arcs due to pairs of blind writes. These can be swapped without modifying the *reads-from* relationships.
Once the graph is acyclic it is possible to find a serial schedule view-equivalent to the initial one.

***How to find out if a schedule is in VSR?***

Do the Conflict Graph.
If such graph is cyclic, consider the cycles that are due to blind writes.
Switch such writes (switch the direction of the arrow).
If now the graph is acyclic we can say it is in VSR.

## CSR - Conflict Serializability

***Conflict***
Two operations $o_i$ and $o_j$ $(i \neq j)$ are in ***conflict*** if they address the same resource and at least one of them is a write.
Thus, the possible conflicts are:

- *read-write* conflicts
- *write-write* conflicts.

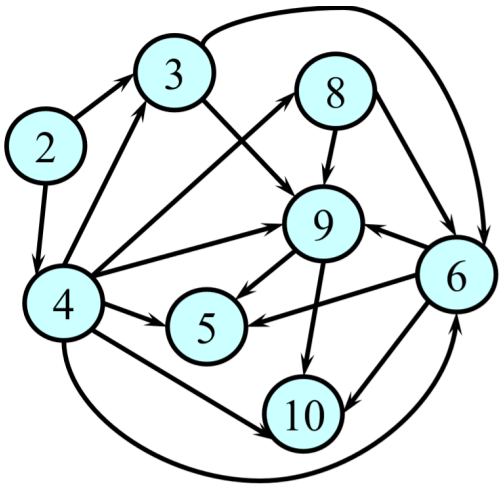
***Conflict-Equivalence***
Two schedules are ***conflict-equivalent*** $(S_i \approx_{\mathbf{C}} S_j)$ if:
$S_i$ and $S_j$ contain the same operations and all conflicting pairs occur in the same order

***Conflict-Serializable***
A schedule is ***conflict-serializable*** if it is conflict-equivalent to a serial schedule of the same transactions. The class of conflict-serializable schedules is named ***CSR***.

***How to find out if a schedule is in CSR?***
We need to do the *Conflict Graph*:

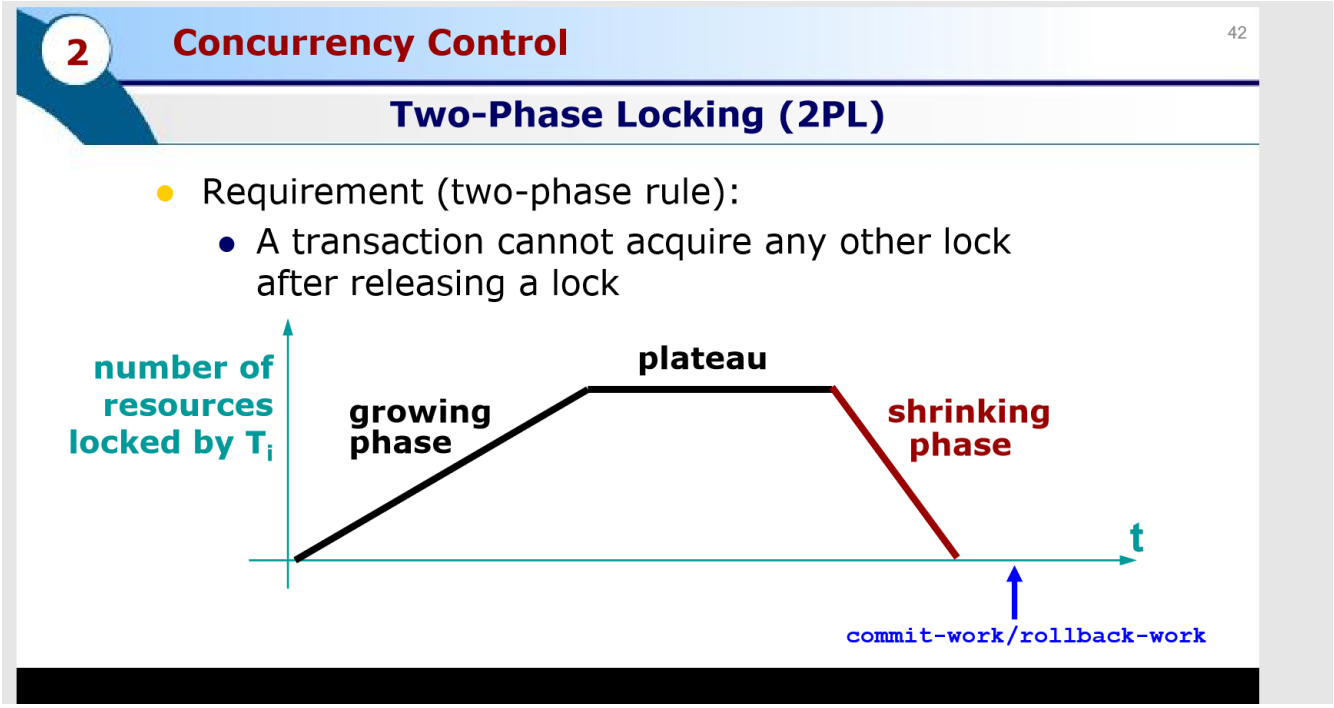
- One node for each transaction $T_i$.
- One arc from $T_i$ to $T_j$ if there exists at least one conflict between an operation $o_i$ of $T_i$ and an operation $o_j$ of $T_j$ such that $o_i$ precedes $o_j$.

A schedule is in ***CSR*** if and only if its conflict graph is acyclic.




Is there any cycle? No.
The schedule is in CSR (and then also in VSR). The graph suggests, e.g., the serial schedule 2,4,3,8,6,9,5,10 as view equivalent (built choosing and deleting nodes without incoming arcs)

# 2PL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Two-Phase Locking (2PL)
Requirement (two-phase rule):
A transaction cannot acquire any other lock after releasing a lock
number of resources locked by $T_i$
growing phase
plateau
shrinking phase
t
commit-work/rollback-work

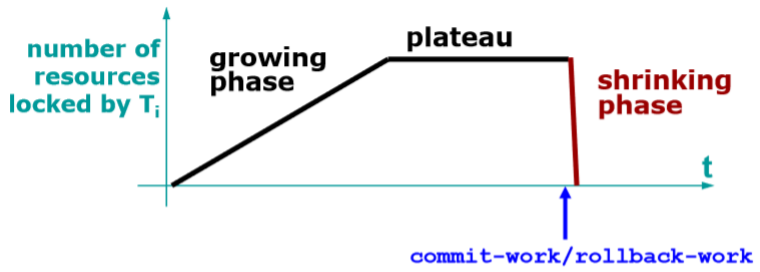
## 2PL STRINCT

## Dirty reads are still a menace: Strict 2PL

- Up to now, we were still using the hypothesis of *commit-projection* (no transaction in the schedule aborts)
  - 2PL, as seen so far, does not protect against dirty reads (and therefore neither do VSR nor CSR)
    - Releasing locks before rollbacks exposes "dirty" data
- To remove this hypothesis, we need to add a constraint to 2PL, that defines **strict 2PL**:
  - *Locks held by a transaction can be released only **after** commit/rollback*
- This version of 2PL is used in commercial DBMSs



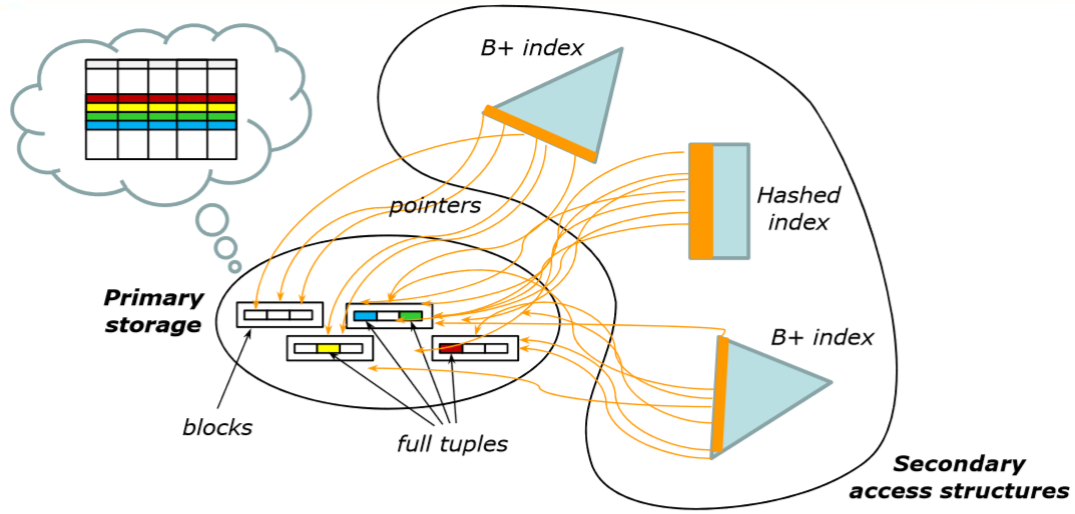
## Strict 2PL in Practice




in 2PL strict, if I unlock the transition 3, I need to unlock it immediately for all the resources.

# Physical Access Structures Introduction

Each table is stored into exactly one primary physical access structure, and may have one or many optional secondary access structures.




- *Primary Structure*
  - Contains all the tuples of a table
  - Its main purpose is to store the table content
- *Secondary Structures*
  - Used to index primary structures, and only contain the values of some fields, interleaved with pointers to the blocks of the primary structure
  - Their main purpose is to speed up the search for specific tuples within the table, according to some search criterion

| | **Primary** | **Secondary** |
|---|---|---|
| **Sequential** structures | **Typical** | - |
| **Hash-based** structures | Frequent | Frequent |
| **Tree-based** structures | Frequent | **Typical** |

# Physical Access Structure Types

Legenda :

- $val(A_j)$ = number of distinct values of the attribute $A_j$ in the table $T$

Three types :

## Sequential Structures

Three types:

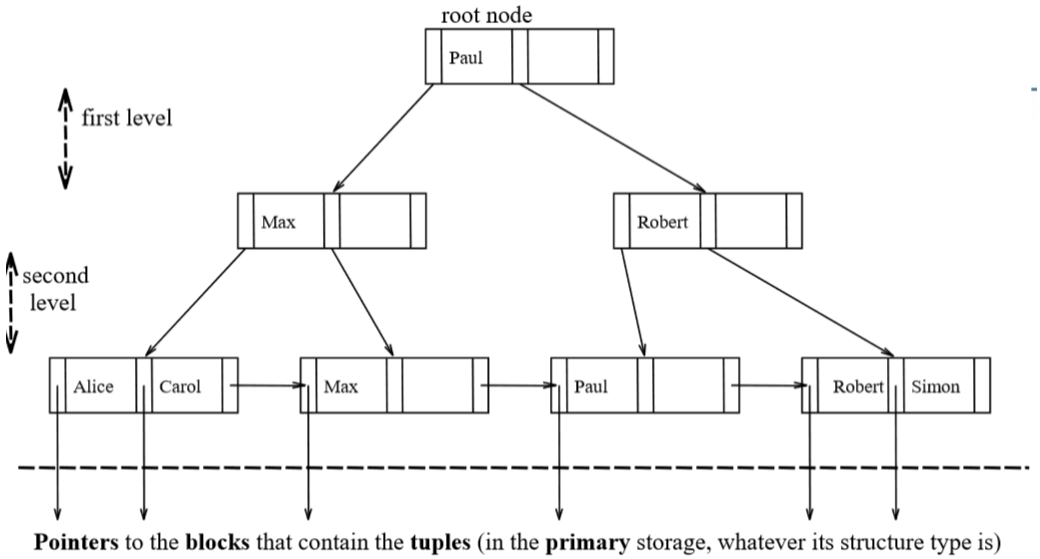
- *Entry - Sequenced*
  - In an Entry - Sequenced organization, the sequence of the tuples is dictated by their order of entry
  - Optimality
    - Optimal for
      1. Space occupancy
      2. Carrying out sequential reading and writing
    - Non - Optimal for
      1. Searching specific data units
      2. Updates that increase the size of tuple, in fact all following tuples must be shifted on.
- *Array*
  - In an array organization, the tuples (all of the same size) are arranged as in an array, and their position depend on the values of an index (or indexes)
  - Appropriate only when the tuples are of fixed length
- *Sequentially - Ordered*
  - In a Sequentially - Ordered organization, the tuples are ordered according o the value of a key (typically one field, but may be obtained by combining more than one attribute)
  - Main problem: Insertions and Updates could increase the data size, this means that such structures require reordering techniques for the tuples already present. There are some techniques to avoid such reordering problem (i.e. leaving a certain number of slots free at the time of first loading, followed by 'local reordering' operations)

## Hash - Based Access Structures

Such structures assure an efficient associative access to data, based on the value of a key field to which the hash is applied. Hash - Based structures have N_B buckets, each typically of the size of 1 block. A hash algorithm is applied to the key field so as to compute a value between $0$ and $N_B - 1$.

- When is it used? This is the most efficient structure for queries with equality predicates, but it is rather inefficient for interval queries

## Tree Structures




Gives associative access based on the value of a key field.

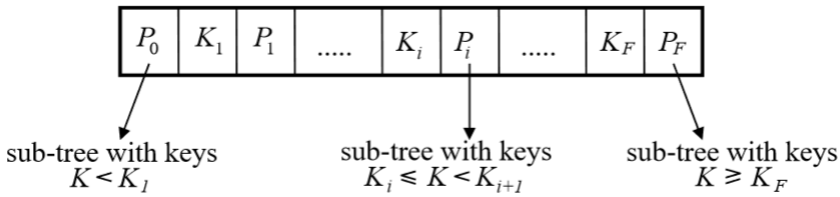
Each tree has:

- One root node
- Several intermediate nodes
- Several leaf nodes

Each node corresponds to a block.

The links between nodes are established by pointers to mass memory.

*Search technique* :




Suppose we are looking for a tuple with key value V, so at each intermediate node we do the following:

- If V < K_1 follow P_0
- if V >= K_F follow P_F
- Otherwise, follow P_J such that K_J <= V <K_j+1

The nodes can be organized in two ways:

- *B+ trees*
  - Pointers to the tuples are only contained in the leaf nodes
  - The leaf nodes are linked in a chain ordered by he key
  - Supports interval queries efficiently
  - The most used by relational DBMSs
- *B trees*
  - Pointers to the tuples can also be contained in intermediate nodes. In this way, for tuples with V = K_i , we don't kneed to descend the tree down to the leaves.
  - No chain links for leaf nodes, namely there is no support for intervals

# Costs

- **Sequential Structures**
  - Cost of Equality Lookups: a full scan, because lookups are no supported
  - Cost of Interval Lookups: a full scan, because lookups are not supported
- **Hash and Tree Structures**
  - Cost of Equality Lookups:
    The lookup is supported only if A_i is the key attribute on which the structure is built, otherwise, a full scan is required. Lookup cost depends on
    - The storage type (primary/secondary)
    - The structure type (tree/hash)
    - The key type (unique/non-unique)

    *If it is a primary hash:*

    A lookup for A_i = v needs to find the block(s) storing the tuples with the same hash for A_i as those with v. The Hash function is applied to v, the right bucket is therefore immediately identified. The bucket contains the searched tuples
    - $Cost = 1$ if the hash has no overflow chains
    - $Cost = 1 + (average\ size\ of\ the\ overflow\ chain)$ if the hash has overflow chains

    *If it is a secondary hash:*
    The lookup for $A_i = v$ works exactly in the same way, only the aim is to find the block(s) storing the pointers to the tuples with the same hash for $A_i$ as those with $v$. The Hash function is applied to $v$, the right bucket is identified, and the corresponding block(s) are retrieved.
    - $Cost = 1$ (for the initial bucket block) +

      ```
      $(average  \ overflow  \ blocks) \\ $ (if  any) • +
      ```

      $(1\ block\ per\ pointer\ to\ be\ followed)$
  - Cost of Interval Lookups:
    lookups based on intervals are not supported
- **B+ Structures**
  - Cost of equality Lookups :
    *If it is a primary B+*
    A lookup for A_i = v needs to find the block(s) storing all tuples indexed by the key value v. the root is read first, then, a node per intermediate level is read, until the first leaf node that stores tuples with A_i = v is reached. If the searched tuples are all stored in that leaf block, stop. Else, continue in the leaf blocks chain until v changes to v'
    - $Cost =$
      $(1\ block\ per\ intermediate\ level) +$
      $(as\ many\ leaf\ blocks\ as\ necessary\ to\ store\ all\ the\ tuples\ with\ A_i = v)$

    *If it is a secondary B+*
    The lookup for $A_i = v$ needs to find the block(s) storing all the pointers that point to all tuples with the key value $v$ for $A_i$. The root is read first, then a node per intermediate level is read, until the first leaf node that sores pointers A_i = v is reached. If the pointers are all stored in that leaf block, stop.

Else, continue in the leaf blocks chain until v changes to $v'$.

- $Cost = (1\ block\ per\ intermediate\ level) + (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ the\ pointers\ pointing\ to\ the\ tuples\ with\ A_i = v) + (1\ block\ per\ each\ pointer)$

- Cost of Interval Lookups

  *If it is a primary B+*
  We consider a lookup for $v_1 \leq A_i \leq v_2$ as the general case. If $(A_i < v)$ or $(v < A_i)$ we just assume that the other edge of the interval is the first/last value in the structure. The root is read first, then a node per intermediate level is read, until the first leaf node that stores tuples with $A_i = v_1$ is reached. if the searched tuples are all stored in that leaf block stop. Else continue in the leaf blocks chain until v_2 is reached.

  - $Cost = (1\ block\ per\ intermediate\ level) + (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ all\ the\ tuples\ in\ the\ interval)$

  *If it is a secondary B+*

  We still consider a lookup for $v_1 \leq A_i \leq v_2$ as the general case. The root is read first, then a node per intermediate level is read, until the first leaf node that stores a the pointers pointing to the tuples with $A_i = v_1$ is reached. If all the pointers (up to $v_2$) are in that leaf block, stop. Else, continue in the leaf blocks chain until $v_2$ is reached.

  - $Cost = (1\ block\ per\ intermediate\ level) + (as\ many\ leaf\ blocks\ as\ necess.\ to\ store\ all\ pointers\ to\ the\ tuples\ in\ theinterval)\ + (1\ block\ per\ each\ such\ pointer,\ in\ order\ to\ retrieve\ the\ tuples)$

## Cost of Joins

I cannot talk about the cost of joins regarding each type of structure independently, that's because joins are to do among different tables, that can be represented in different structures one from the other.

Legenda:

- $b_A$ = Number of blocks of the table A
- $t_A$ = Number of tuples of a block of the table A
- $E$ = External table
- $I$ = Internal table
- $L$ = Left
- $R$ = Right

Type of joins:

- *Nested - Loop Join Cost (Assuming not enough free pages to host an entire table in the buffer)*

  compares the tuples of a block of table T_E with all the tuples of all the blocks of T_I before moving to the next block T_E.

  - $Cost = b_E * b_I$

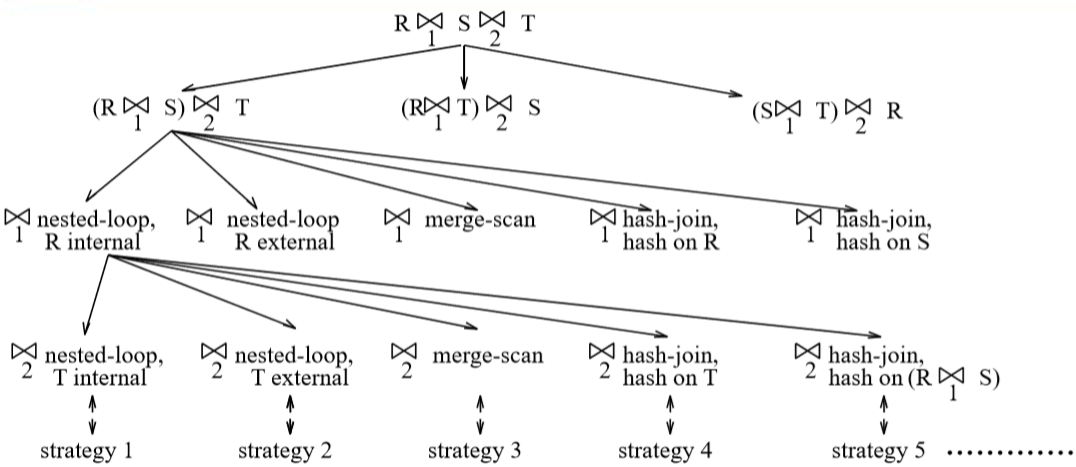
- *Nested - Loop Join Cost (Assuming enough free pages to host an entire table in the buffer)*
  the table in the buffer is chosen as internal table.
  - $Cost = b_E + b_I$
- *Nested - Loop Join Cost (Assuming that one table supports indexed access in the form of a lookup based on the join predicate)*
  such table is chosen as internal
  - $Cost = b_E + (t_E * (cost\ of\ one\ indexed\ access\ to\ T_I)$
- *Merge - Scan Join Cost (Assuming the primary storage to be sequentially ordered or B+)*
  This join is possible only if both tables are ordered according to the same key attribute, that is also the attribute used in the join predicate. The cost is linear in the size of the table.
  - $Cost = b_L + b_R$
- *Hashed Joins (Assuming the two hashes to be both primary storages)*
  This join is possible only if bot tables are hashed according to the same key attribute, that is the same used in the join predicate. The cost is linear in the size of the hashes
  - $Cost = b_L + b_R$

# Query Planning

What cost to assign to a plan?

- Construct a decision tree, in which each node corresponds to a choice, each leaf node corresponds to a specific execution plan
- Assign to each plan a cost

$$C_{total} = C_{I/O} n_{I/O} + C_{cpu} n_{cpu}$$

- Choose the plan with the lowest cost, based on operations research (branch and bound)

Example of decision tree: